Notes: Bazzi, Hilmy, and Marx (RES, 2026)

Religion, Education, and the State

Contents

1 Big picture

- **Question.** How do state and religious providers of education compete during nation building? The paper studies whether a mass expansion of state schooling crowds out religious schooling or instead stimulates a parallel religious education system.
- **Historical setting.** Indonesia implemented the SD INPRES school construction programme in the 1970s, one of the largest public primary school expansions in history. The programme was part developmental policy and part nation-building policy under Suharto's secular-nationalist regime.
- **Competing education systems.** Indonesia had a dual system: secular public schools and Islamic schools. Islamic schools included formal madrasa at primary, junior secondary, and senior secondary levels, plus informal religious schools such as pesantren and diniyah.
- **Core empirical idea.** The programme allocated more state schools to areas with more children not enrolled in school. Many such areas had pre-existing Islamic schooling. This created variation in the intensity of state schooling across districts and villages.
- **Main school-supply result.** Rather than simply crowding out Islamic education, SD INPRES stimulated entry of Islamic schools, especially formal secondary madrasa and informal religious schools. The Islamic sector responded dynamically to the state expansion.
- **Main mechanism.** State primary schools increased demand for post-primary schooling. Islamic schools entered or expanded to absorb this demand, especially in villages with stronger funding capacity from rice wealth and waqf endowments.
- **Curriculum result.** New Islamic schools in high-INPRES districts did not become more secular to compete with public schools. Instead, many became more differentiated, devoting more curriculum time to Islamic subjects and Arabic.
- **School-choice result.** Exposed cohorts shifted away from elementary madrasa and toward secondary madrasa. SD INPRES therefore reduced Islamic primary attendance but increased Islamic secondary education.
- **Political result.** SD INPRES did not increase electoral support for Suharto's regime. In some specifications, support for the secular regime weakened in high-INPRES markets.
- **Identity result.** Exposed cohorts show stronger religious identity and practices: greater Arabic literacy and stronger religious piety, but not stronger national identity.
- **Intergenerational result.** Religious identity effects transmit within households. The children of exposed cohorts are more likely to receive Arabic in the home and to have parents or partners connected to Islamic education.
- **Economic takeaway.** State-building through mass public goods provision can unintentionally strengthen parallel non-state provision when local organizations can mobilize resources and differentiate their product.

2 Institutional background and conceptual framework

2.1 The dual education system

- Indonesia's education system combined secular state schools with a historically important Islamic education sector.
- Islamic schools were heterogeneous. Formal madrasa offered secular subjects plus Islamic instruction and were organized by level: elementary, junior secondary, and senior secondary.
- Informal Islamic schools included pesantren, often residential and centered on religious instruction, and diniyah, which typically provided supplemental Islamic education.

- Islamic schools were often locally financed through donations, waqf endowments, community resources, and informal taxation.

Interpretation: The religious education sector was not merely a set of private schools. It was a decentralized local public-goods system, embedded in religious institutions and community finance.

2.2 The SD INPRES programme

- SD INPRES was a mass construction programme for public primary schools launched in the 1970s.
- The allocation rule targeted districts with more children not enrolled in school in 1971.
- The programme was meant to expand access to public primary education and to promote national development and national identity.
- Because many under-enrolled areas had Islamic schools, the programme created direct competition between the secular state and religious providers.

Interpretation: The allocation rule produces an empirical source of variation: areas with greater unmet schooling demand received more public primary schools. The authors control flexibly for the original allocation variables to isolate treatment variation.

2.3 Predicted margins of response

The paper studies three connected margins:

1. **Religious school entry.** Islamic schools may exit, fail to enter, or enter strategically in response to state-school expansion.
2. **Religious school choice.** Families may substitute from Islamic primary schools into public primary schools, but public primary graduates may later choose Islamic secondary schools.
3. **Political and cultural identity.** Public schooling may promote national identity, but it may also trigger religious counter-mobilization if secular schooling is viewed as threatening.

Interpretation: These margins are linked. State primary expansion can crowd out Islamic primary schools while simultaneously increasing demand for Islamic secondary schools.

3 Data and measurement

3.1 Public school construction

- The treatment is SD INPRES school construction from 1973 to 1978.
- At the district level, treatment intensity is the number of SD INPRES schools built per 1,000 school-aged children in 1971.
- At the village level, treatment is a binary indicator for whether an SD INPRES school entered the village.
- The paper uses both continuous treatment intensity and discretized high-vs-low exposure for synthetic difference-in-differences.

Interpretation: The same reform can be studied at different spatial scales: district variation captures broad exposure, while village entry identifies local market responses around actual public-school construction.

3.2 Islamic school registries

- The paper assembles administrative data on Islamic school entry and school type.
- Outcomes include new schools per district-year or village-year, separated into elementary madrasa, junior secondary madrasa, senior secondary madrasa, pesantren, diniyah, and all Islamic schools.
- The registry allows the authors to study dynamic entry over time, not just cross-sectional school counts.

Interpretation: Entry timing is essential because the theoretical question is whether Islamic schools responded to state school construction rather than merely being correlated with it.

3.3 Curriculum and formalization

- For some Islamic schools, the paper observes curriculum information and school names.
- Curriculum outcomes include time allocated to Islamic subjects, Arabic, Indonesian, and Pancasila/civic education.
- The paper also studies the share of entering schools that are formal madrasa versus informal religious schools.

Interpretation: Curriculum outcomes distinguish two possible strategic responses: religious schools could secularize to compete with the state, or differentiate by offering more religious content.

3.4 School choice data

- School-choice outcomes come from annual Susenas household survey data from 2012 to 2018.
- Outcomes include whether the individual's highest completed education level was elementary madrasa, junior secondary madrasa, senior secondary madrasa, or any madrasa.
- Exposure is based on district of birth and age in 1974.
- Young cohorts are those aged 2–6 in 1974, who were fully exposed to SD INPRES when entering primary school.
- Older cohorts aged 12–17 in 1974 serve as comparison cohorts in the baseline DID.

Interpretation: This design asks how childhood exposure to public primary school expansion changed completed religious schooling many years later.

3.5 Political, identity, and transmission outcomes

- Electoral outcomes include district-level vote shares for Suharto's party and for Islamic parties across legislative elections.
- Identity outcomes include national language use at home, Arabic literacy, and religious practices such as prayer, fasting, Qur'an reading, and zakat.
- Intergenerational outcomes examine whether exposed parents transmit Arabic and Islamic schooling preferences to children.

Interpretation: These outcomes allow the paper to move from schooling markets to nation-building and long-run cultural consequences.

4 Empirical specifications

4.1 District-level school-entry DID

The baseline district-year specification is:

$$y_{jt} = \theta_j + \theta_t + \beta(\text{INPRES}_j \times \text{post1972}_t) + (X_j'\theta_t)'\eta + \varepsilon_{jt}. \tag{1}$$

- y_{jt} is new Islamic schools of a given type per district-year and per 1,000 children.
- INPRES_j is SD INPRES construction intensity from 1973 to 1978 per 1,000 children in 1971.
- post1972_t equals one for years after the onset of the programme.
- θ_j and θ_t are district and year fixed effects.
- $X_j'\theta_t$ allows predetermined district characteristics to have flexible time-varying effects.

Interpretation: The coefficient β measures whether higher-INPRES districts experienced greater Islamic school entry after the public-school expansion.

4.2 Synthetic difference-in-differences

The paper also estimates synthetic difference-in-differences using a binary high-INPRES treatment:

$$HighINPRES_j = \mathbf{1}\{\text{INPRES}_j > 51\text{st percentile}\}. \tag{2}$$

Interpretation: SDID creates a weighted comparison group that better matches pre-treatment outcomes in high-INPRES districts. It is especially useful for visualizing dynamic responses.

4.3 Village-level event-study design

The village-year design exploits staggered local entry of SD INPRES schools:

$$y_{vt} = \theta_v + \theta_t + \sum_{\tau} \gamma_\tau \text{INPRES}_{v,t-\tau} + (X_v'\theta_t)\eta + \varepsilon_{vt}. \tag{3}$$

- $\text{INPRES}_{v,t-\tau}$ indexes event time relative to entry of an SD INPRES school in village v.
- The design studies whether Islamic schools enter immediately before or after state schools enter the village.
- The paper also uses the robust and efficient estimator of Borusyak, Jaravel, and Spiess for staggered treatment timing.

Interpretation: The village design is closer to a local market shock: did Islamic schools enter the same village market after a public school entered?

4.4 Funding heterogeneity specification

To study financing of Islamic school entry, the paper estimates:

$$\begin{aligned} y_{vt} = \theta_v + \theta_t + \beta_0\text{INPRES}_{vt} + \beta_1(\text{INPRES}_{vt} \times RiceYield_{v0}) \quad (4)\\ + \beta_2(\text{INPRES}_{vt} \times Waqf_{v0}) + (X_v'\theta_t)\eta + \varepsilon_{vt}. \quad (5) \end{aligned}$$

- $RiceYield_{v0}$ proxies for local capacity to raise informal taxes or donations from agricultural surplus.
- $Waqf_{v0}$ captures the presence of Islamic endowments that can fund religious public goods.
- The key coefficients are β_1 and β_2.

Interpretation: A stronger response in high-rice-yield or high-waqf villages supports the interpretation that community financing helped sustain Islamic school entry.

4.5 Curriculum differentiation specification

For Islamic school curriculum, the paper estimates an unbalanced school-grade-year panel analogue of the district DID:

$$Curriculum_{sjt} = \delta_s + \delta_j + \delta_t + \beta(\text{INPRES}_j \times \text{post1972}_t) + \Gamma X_{jt} + \varepsilon_{sjt}. \tag{6}$$

Interpretation: The goal is to compare schools entering before and after SD INPRES across areas with different INPRES intensity. A positive coefficient on Islamic subject share means schools in high-INPRES districts became more religiously differentiated.

4.6 School-choice exposure design

The cohort-level exposure specification is:

$$y_{ijt} = \theta_j + \theta_t + \beta(\text{INPRES}_j \times \text{young}_{it}) + (X_j'\theta_t)'\eta + \varepsilon_{ijt}. \tag{7}$$

- i indexes individuals, j district of birth, and t cohort or survey year.
- young_{it} equals one for cohorts aged 2–6 in 1974.
- Outcomes are indicators for completing different levels of madrasa schooling.

Interpretation: The coefficient β measures how exposure to public primary school expansion changed the religious-schooling path of affected children.

4.7 Election and identity specifications

The electoral and identity analyses use analogous exposure designs:

- for elections, β is allowed to vary by election year;
- for identity and religiosity, the outcomes are language use, Arabic literacy, and religious practices;
- for intergenerational transmission, the exposure is assigned to parents and linked to child outcomes.

Interpretation: These specifications ask whether schooling-market responses translated into broader political and cultural consequences.

5 Identification and empirical design

5.1 Why the SD INPRES allocation rule helps

- The programme targeted areas with many unenrolled school-aged children.
- The paper controls flexibly for the policy-rule variables, including 1971 child population and school enrollment.
- District and village fixed effects absorb time-invariant differences across locations.
- Year fixed effects absorb national trends in school entry and identity outcomes.

Interpretation: Identification relies on comparing places with different INPRES intensity after controlling for the determinants of allocation and fixed location characteristics.

5.2 Pre-trends and dynamic evidence

- Figures 2–4 show limited pre-trends in Islamic school entry before the programme.
- SDID estimates match high- and low-INPRES trends before treatment and compare post-treatment divergence.
- Village event studies show entry after local SD INPRES construction.

Interpretation: The graphical and event-study evidence supports the interpretation that the Islamic sector response followed public school entry rather than preceding it.

5.3 Mechanism evidence

- Funding heterogeneity shows stronger entry in villages with high potential rice yields and larger waqf endowments.
- Curriculum evidence shows differentiation rather than secularization.
- Formalization evidence shows formal madrasa gained relative importance even as informal Islamic schooling also expanded.
- School-choice evidence shows public primary expansion increased demand for Islamic secondary schooling.

Interpretation: The evidence points to a strategic response by religious providers: mobilize local resources, expand supply, and differentiate on religion.

5.4 Threats and robustness

- The main threat is that high-INPRES areas may have had different trends in Islamic schooling even without the programme.
- The paper addresses this using pre-trend checks, SDID, robust staggered treatment estimators, and controls for religious and political baseline variables.
- Another concern is survivor bias in school registries; the authors use robustness checks and multiple school types to address this.
- For schooling outcomes, selection into further education is addressed using conditional outcomes and selection corrections.

Interpretation: The empirical case is strongest because multiple independent margins – entry, curriculum, school choice, elections, identity – point in a coherent direction.

6 Tables and figures

6.1 Figure 1 and Table 1: targeting of SD INPRES

Read:

- Figure 1 shows that SD INPRES construction followed the official policy rule: more schools were built where more children were not enrolled.
- The figure also shows that high-INPRES places had more pre-existing Islamic schooling.
- Table 1 confirms these patterns in regressions. At the district level, Islamic primary enrollment and the stock of Islamic primary schools predict more INPRES construction.
- At the village level, SD INPRES was more likely to enter villages with private Islamic elementary schools and less likely to enter villages with private non-Islamic elementary schools.

Economic message: The state-school expansion directly intersected with Islamic education markets. This creates the competitive setting studied in the paper.

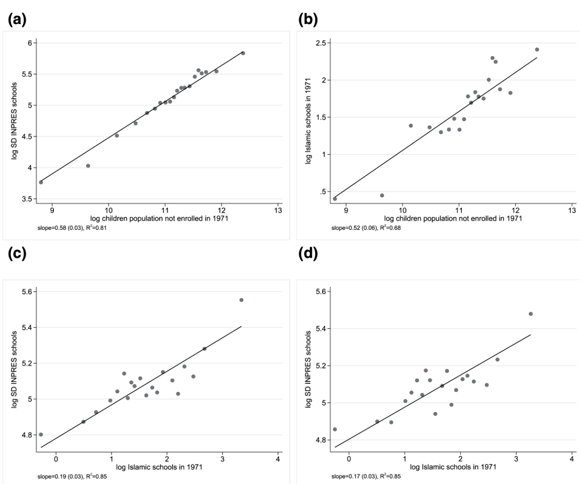

Figure 1: targeting of INPRES construction

TABLE 1

Correlates of INPRES elementary school allocation

	Dependent variable:				
	log SD INPRES in district				SD INPRES in village
District Level	(1)	(2)	(3)	(4)	(5)
% Islamic primary enrollment, 1967–72	0.039***	0.028***		0.011	
	(0.009)	(0.009)		(0.008)	
Log school-aged children not enrolled, 1971	0.684***		0.622***	0.628***	
	(0.076)		(0.080)	(0.072)	
% Non-Islamic primary enrollment, 1967–72		−0.016***		−0.014***	
		(0.005)		(0.005)	
Log Islamic primary schools, 1971			0.130***	0.079***	
			(0.030)	(0.025)	
Islamic parties vote share, 1950s				0.004***	
				(0.001)	
Village Level					
Any public elementary in villages, 1971					−0.028**
					(0.012)
Any private non-Islamic elementary in villages, 1971					−0.046***
					(0.015)
Aany private Islamic elementary in villages, 1971					0.052***
					(0.019)
Number of districts or villages	275	275	275	275	75,208
Targeting policy controls	✓	✓	✓	✓	✓
R^2	0.872	0.812	0.872	0.893	0.030

Notes: This table reports correlates of SD INPRES school construction at the district and village levels. The dependent variable is the log number of INPRES elementary schools constructed at the district level between 1973 and 1978 (columns 1–4) and an indicator for any SD INPRES built in the village during that same period (column 5). All regressions control for the variables that informed the policy rule for INPRES school allocations: province fixed effects, the 1971 children population, the 1971 enrollment rate, and exposure to the water and sanitation programme. $^{*} p < 0.1$, $^{**} p < 0.05$, $^{***} p < 0.01$. Robust standard errors in parentheses, clustered by district in column 5.

Table 1: correlates of INPRES allocation

6.2 Table 2 and Figures 2–4: Islamic school entry

Read:

- Table 2 shows greater Islamic school entry in high-INPRES districts and villages.
- Panel (a) DID estimates show positive effects on all categories of Islamic schools, including elementary madrasa, junior secondary madrasa, senior secondary madrasa, pesantren, and diniyah.
- The effect on all Islamic schools is about 0.0104 new schools per district-year per 1,000 children in the district DID.
- The SDID estimate in panel (b) is larger, about 0.0176 for all Islamic schools.
- Village-level DID and robust DID estimates show similar positive entry responses around local SD INPRES entry.
- Figures 2, 3, and 4 show dynamic evidence: high-INPRES areas diverge after the programme, with limited evidence of pre-trends.

Economic message: Public primary school expansion did not crowd out Islamic education supply. Instead, it stimulated entry by religious providers, especially in markets where public schooling created new demand for continuing education.

TABLE 2
SD INPRES intensity and islamic school entry

	Formal *Madrasa*			Informal		All
	Elementary (1)	Junior sec. (2)	Senior sec. (3)	*Pesantren* (4)	*Diniyah* (5)	Islamic (6)
	(a) Difference-in-differences, district level					
INPRES × post-1972	0.0017***	0.0016***	0.0009***	0.0021***	0.0041**	0.0104***
	(0.0005)	(0.0004)	(0.0002)	(0.0005)	(0.0016)	(0.0023)
	(b) Synthetic difference-in-differences, district level					
INPRES × post-1972	0.0034***	0.0034***	0.0015***	0.0041***	0.0053*	0.0176***
	(0.0013)	(0.0009)	(0.0004)	(0.0011)	(0.0027)	(0.0039)
1959 Islamic schools × year FE	✓	✓	✓	✓	✓	✓
District FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Number of district–years	11,000	11,000	11,000	11,000	11,000	11,000
Dep. var. mean	0.007	0.005	0.002	0.007	0.018	0.039
R^2 (panel (a))	0.178	0.169	0.166	0.313	0.565	0.463
	(c) Difference-in-Differences, Village Level					
SD INPRES Entry	0.0021***	0.0018***	0.0007***	0.0017***	0.0043***	0.0105***
	(0.0003)	(0.0002)	(0.0001)	(0.0003)	(0.0007)	(0.0011)
	(d) Robust Difference-in-Differences Estimator, Village Level					
SD INPRES Entry	0.0022***	0.0017***	0.0008***	0.0013***	0.0035***	0.0094***
	(0.0002)	(0.0001)	(0.0001)	(0.0002)	(0.0003)	(0.0005)
1959 Islamic schools × year FE	✓	✓	✓	✓	✓	✓
Village FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Number of village–years	3,334,560	3,334,560	3,334,560	3,334,560	3,334,560	3,334,560
Dep. var. mean	0.0009	0.0001	0.0001	0.0006	0.0025	0.0041
R^2 (panel (c))	0.035	0.029	0.029	0.068	0.063	0.075

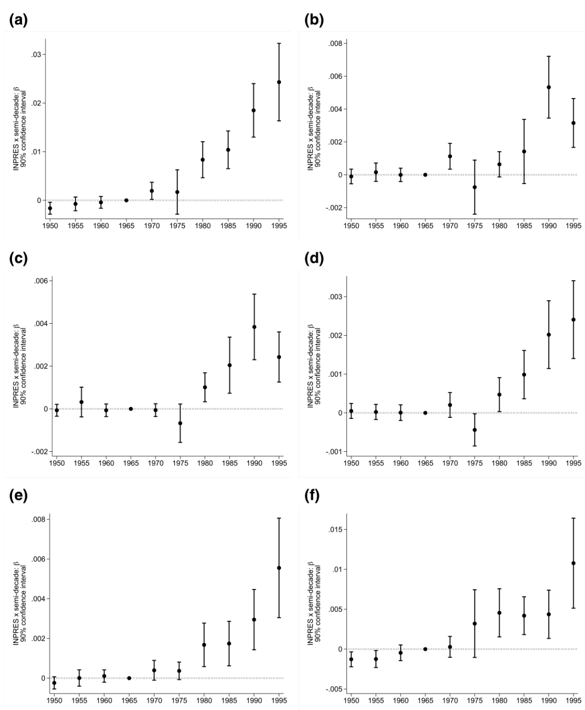

Figure 2: district DID event-study estimates

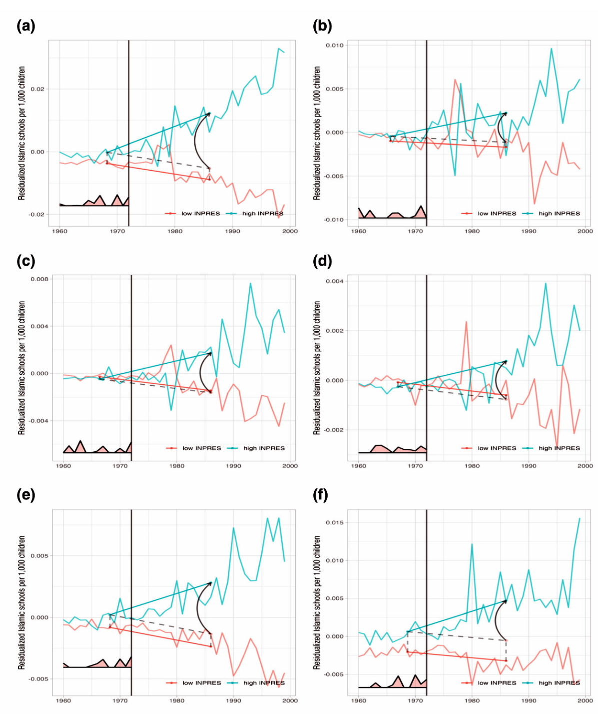

Figure 3: synthetic DID entry estimates

6.3 Table 3: financing Islamic school entry

Read:

- Table 3 interacts SD INPRES entry with local funding-capacity proxies.
- Villages with higher potential rice yields show stronger entry of formal madrasa and all Islamic schools.
- Villages with larger waqf endowments show stronger entry of informal Islamic schools, especially pesantren

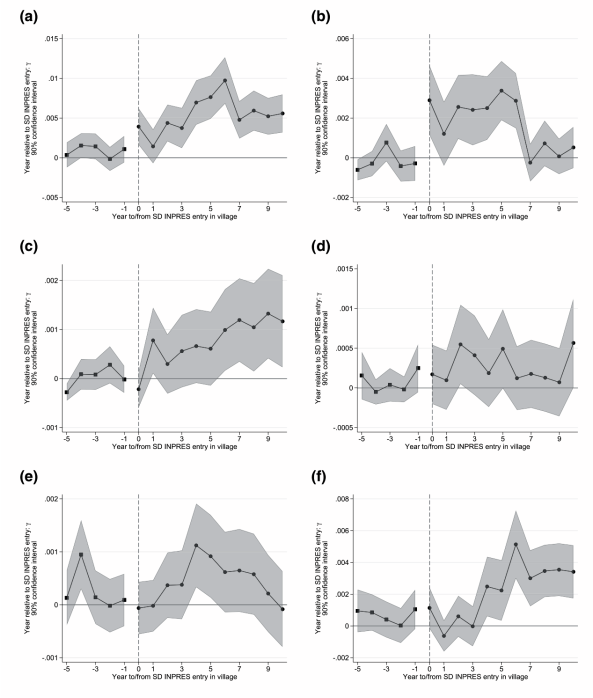

TABLE 3

Funding of islamic school entry, village level

	Formal *Madrasa*		Informal		All
	Elementary (1)	Secondary (2)	*Pesantren* (3)	*Diniyah* (4)	Islamic (5)
SD INPRES	0.0007***	0.0012***	−0.0002	−0.0001	0.0016***
	(0.0002)	(0.0002)	(0.0001)	(0.0002)	(0.0004)
SD × high potential rice yield	0.0005	0.0008***	0.0003	0.0016***	0.0032***
	(0.0003)	(0.0002)	(0.0003)	(0.0005)	(0.0007)
SD × high *waqf* endowment	−0.0003	−0.0004	0.0015***	0.0025***	0.0033***
	(0.0003)	(0.0002)	(0.0002)	(0.0004)	(0.0006)
SD villages with high funding base	0.0009***	0.0016***	0.0016***	0.0040***	0.0081***
	(0.0003)	(0.0002)	(0.0003)	(0.0004)	(0.0006)
Village FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Year FE × …	✓	✓	✓	✓	✓
Islamic schools in 1959	✓	✓	✓	✓	✓
Secular primary schools in 1959	✓	✓	✓	✓	✓
high potential rice yield	✓	✓	✓	✓	✓
high *waqf* endowment	✓	✓	✓	✓	✓
Number of village-years	3,007,920	3,007,920	3,007,920	3,007,920	3,007,920
Dep. var. mean	0.0009	0.0001	0.0005	0.0029	0.0045
R^2	0.041	0.037	0.068	0.064	0.077

and diniyah.

- The triple high-funding-base interaction shows strong responses across school types.

Economic message: Islamic school expansion required resources. The stronger response in rice-rich and waqf-rich villages supports a community-mobilization mechanism.

6.4 Table 4 and Figure 5: curriculum differentiation and formalization

Read:

- Table 4 shows that Islamic schools created after SD INPRES in high-INPRES districts allocated more time to Islamic subjects.
- The Islamic subject share increases by about 1.2 percentage points overall, and the effect is strongest for junior secondary madrasa.
- Arabic instruction also rises, while Indonesian and civic/Pancasila content do not rise in the same way.
- Figure 5 shows formal and informal Islamic schools as a share of all school entry.
- Formal madrasa expanded rapidly in high-INPRES districts, while informal Islamic schools also retained an important role.

Economic message: Religious schools did not respond to state competition by becoming more secular. They often differentiated by emphasizing religious content, while formal madrasa expanded along the state-like schooling ladder.

TABLE 4

SD INPRES intensity and curriculum differentiation in Islamic schools

	All levels (1)	Primary (2)	Jun. sec. (3)	Sen. sec. (4)
	(a) Islamic subject share			
INPRES × post-1972	0.012*	0.013**	0.022***	−0.040
	(0.007)	(0.006)	(0.008)	(0.024)
Dep. var. mean	0.246	0.238	0.261	0.242
Dep. var. std. dev.	0.047	0.033	0.028	0.036
Number of observations	4,243	1,404	1,662	1,046
Number of districts	258	213	213	178
	(b) Arabic share			
INPRES × post-1972	0.002*	0.003**	0.008***	0.017***
	(0.001)	(0.001)	(0.002)	(0.002)
Dep. var. mean	0.053	0.050	0.068	0.054
Dep. var. std. dev.	0.013	0.009	0.010	0.007
Number of observations	4,243	1,404	1,662	1,046
Number of districts	258	213	213	178
	(c) *Pancasila*/Civic share			
INPRES × post-1972	−0.002	n/a	−0.003	0.008***
	(0.002)		(0.003)	(0.002)
Dep. var. mean	0.052		0.060	0.039
Dep. var. std. dev.	0.012		0.008	0.004
Number of observations	2,775		1,662	1,046
Number of districts	237		213	178
	(d) *Bahasa* Indonesia share			
INPRES × post-1972	−0.004**	0.000	−0.005	0.002
	(0.002)	(0.003)	(0.005)	(0.002)
Dep. var. mean	0.027	0.001	0.123	0.084
Dep. var. std. dev.	0.047	0.008	0.016	0.008
Number of observations	4,243	1,404	1,662	1,046
Number of districts	258	213	213	178
District FE	✓	✓	✓	✓
Grade-level FE	✓	✓	✓	✓
Year-of-entry FE	✓	✓	✓	✓

Table 4: curriculum differentiation

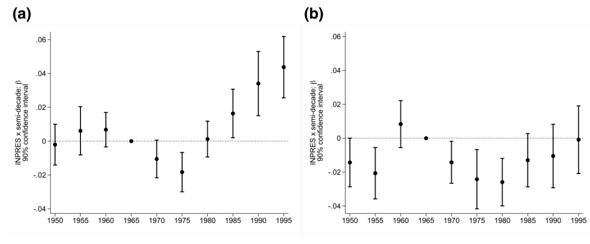

Figure 5: formal and informal Islamic school entry

6.5 Table 5 and Figures 6–7: religious school choice

Read:

- Table 5 shows that childhood exposure to SD INPRES reduces elementary madrasa completion but increases junior and senior secondary madrasa completion.
- In panel (a), the DID estimate for elementary madrasa is negative, while junior and senior secondary estimates are positive and significant.
- The any-level madrasa effect is positive, implying that the secondary increase outweighs the primary decline.
- Figure 6 shows raw cohort patterns: high-INPRES areas shift toward Islamic secondary schooling for exposed cohorts.
- Figure 7 shows estimated cohort effects, separating short-run and long-run exposure patterns.

Economic message: SD INPRES shifted the timing of religious education. Public primary schooling reduced Islamic primary attendance but created more graduates who later entered Islamic secondary schools.

TABLE 5

SD INPRES exposure and Islamic school choice

	Highest education level: [...] *Madrasa*			
	Elementary (1)	Junior secondary (2)	Senior secondary (3)	Any level (4)
	(a) Difference-in-differences			
INPRES × young	−0.0011**	0.0019***	0.0010***	0.0017**
	(0.0004)	(0.0004)	(0.0003)	(0.0007)
	(b) Synthetic Difference-in-Differences			
INPRES × young	−0.0025***	0.0034***	0.0020***	0.0026*
	(0.0008)	(0.0008)	(0.0007)	(0.0014)
District × Survey Year FE	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓
1957 Islamic Schools × Cohort FE	✓	✓	✓	✓
Number of Individuals	839,026	839,026	839,026	839,026
Number of Districts	275	275	275	275
Dep. Var. Mean	0.014	0.011	0.008	0.031
R^2 (panel a)	0.031	0.014	0.009	0.033

Figure 6: raw schooling cohort patterns

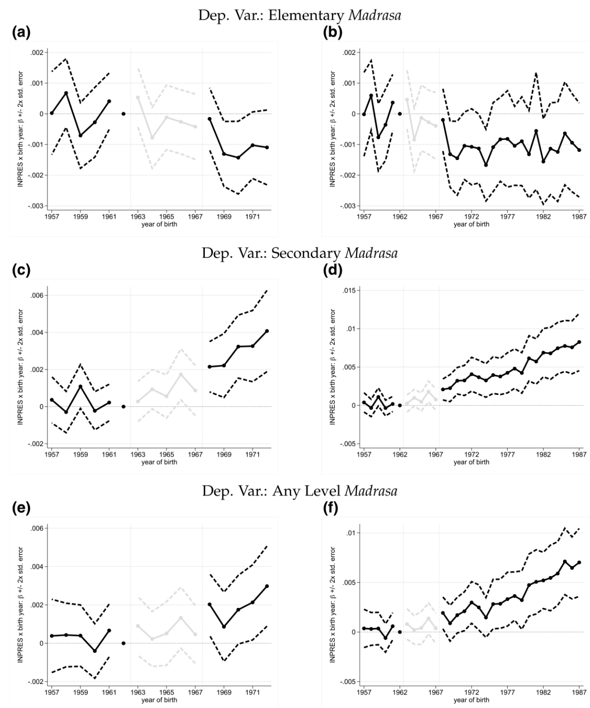

Figure 7: cohort-specific treatment effects

6.6 Tables 6–7: conditional schooling and gender heterogeneity

Read:

- Table 6 shows that the shift toward Islamic secondary schooling remains when outcomes are defined conditional on graduating from a given level.
- This helps address the concern that effects are driven only by higher total years of schooling.
- Selection-corrected specifications are broadly consistent with the unadjusted effects.
- Table 7 shows that SD INPRES increased years of schooling, especially for boys, but also interacted with gender and the later veiling-ban environment.
- Boys of exposed cohorts were less likely to stop at elementary madrasa and more likely to attend secondary madrasa.

Economic message: The school-choice results are not just a by-product of more schooling. Conditional estimates show a real reallocation toward Islamic secondary education.

TABLE 6

SD INPRES exposure and school choice, conditional estimates

	Highest education level: [...] *Madrasa* | Graduating at that Level			
	Elementary (1)	Junior secondary (2)	Senior secondary (3)	Any level (4)
	(a) Difference-in-differences (DID)			
INPRES × young	-0.0017^{**}	0.0057^{***}	-0.0000	0.0011
	(0.0007)	(0.0020)	(0.0014)	(0.0007)
	(b) Synthetic difference-in-differences			
INPRES × young	-0.0043^{***}	0.0117^{***}	-0.0003	0.0016
	(0.0016)	(0.0044)	(0.0028)	(0.0017)
	(c) DID with Selection Correction (Parametric)			
INPRES × young	-0.0031^{**}	0.0040	0.0020	0.0003
	(0.0012)	(0.0027)	(0.0022)	(0.0009)
	[0.049]	[0.347]	[0.481]	[0.731]
Selection terms, p-value	0.197	0.429	0.284	0.261
	(d) DID with Selection correction (semiparametric)			
INPRES × young	-0.0018^{**}	0.0055^{***}	0.0001	0.0005
	(0.0008)	(0.0021)	(0.0015)	(0.0009)
	[0.001]	[0.001]	[0.986]	[0.479]
Selection terms, *p*-value	0.928	0.001	0.413	0.386
District × survey year FE	✓	✓	✓	✓
Cohort FE	✓	✓	✓	✓
1957 Islamic schools × Cohort FE	✓	✓	✓	✓
Number of individuals (panel (a))	283,359	100,874	130,546	514,784
Number of districts	275	275	275	275
Dep. var. mean	0.024	0.070	0.044	0.038

Table 6: conditional school-choice estimates

TABLE 7

Medium-run effects and heterogeneity by gender

	Years of School (1)	Highest Education Level: [...] *Madrasa* Elementary (2)	Secondary (3)	Any (4)
INPRES × born 1968–72	0.1766^{***}	-0.0015^{***}	0.0029^{***}	0.0014^{*}
	(0.0309)	(0.0005)	(0.0007)	(0.0007)
INPRES × born 1973–87	0.2058^{***}	-0.0017^{***}	0.0053^{***}	0.0035^{***}
	(0.0398)	(0.0006)	(0.0012)	(0.0012)
INPRES × born 1968–72 × female	-0.1002^{***}	0.0006	0.0005	0.0010
completed primary before the 1982 veiling ban	(0.0285)	(0.0005)	(0.0006)	(0.0007)
INPRES × born 1973–87 × female	0.0188	0.0007	0.0016^{**}	0.0022^{**}
in school when the veiling ban came into effect	(0.0397)	(0.0007)	(0.0007)	(0.0009)
Number of Individuals	2,315,933	2,315,949	2,315,949	2,315,949
Number of Districts	275	275	275	275
Dep. Var. Mean	8.424	0.011	0.028	0.038
R^2	0.189	0.025	0.029	0.039

Table 7: medium-run effects and gender heterogeneity

6.7 Figure 8: electoral impacts

Read:

- Figure 8 estimates election-year-specific effects of SD INPRES on support for Golkar, the Suharto regime party.
- The programme did not increase support for the secular regime.
- Some panels suggest weaker regime support in high-INPRES districts, consistent with religious counter-mobilization.

Economic message: Public schooling did not automatically produce political loyalty to the state. Where it competed with religious institutions, it may have strengthened opposition or reduced support for the regime.

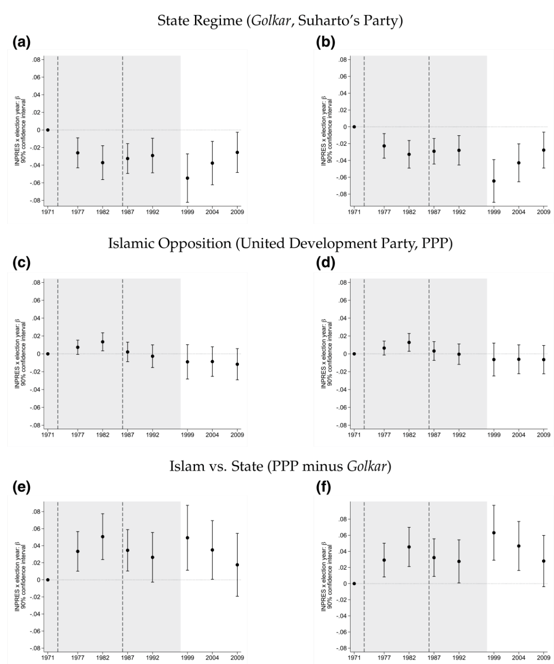

TABLE 8

SD INPRES exposure, identity, and religiosity

	(a) Identity, proxied by language					
	National language use at home			Arabic Literacy		
	All	Muslims	Non-Muslims	All	Islamic-Educated	Secular-Educated
	(1)	(2)	(3)	(4)	(5)	(6)
INPRES × young	-0.0011	-0.0030^{*}	-0.0019	0.0113^{***}	0.0147	0.0024
	(0.0015)	(0.0017)	(0.0021)	(0.0026)	(0.0108)	(0.0025)
Number of individuals	31,680,947	27,811,517	3,869,430	839,026	25,819	813,087
Number of districts	273	273	273	275	264	275
Dep. var. mean	0.166	0.150	0.275	0.343	0.689	0.332

	(b) Islamic piety and practice							
	Pray 5× daily	Fast during Ramadan	Reads the Qur'an	Prayer: Friday	*Sunna*	Group	Pay *Zakat*	Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
INPRES × young	0.1368^{**}	-0.0030	0.0990^{**}	0.1562^{**}	0.0953^{*}	0.0364	0.0357	0.0792^{***}
	(0.0606)	(0.0501)	(0.0481)	(0.0619)	(0.0489)	(0.0465)	(0.0456)	(0.0296)
Number of individuals	1,282	1,283	1,281	1,276	1,268	1,280	1,281	1,284
Number of districts	144	144	144	144	144	144	144	144
Dep. var. mean	0.623	0.797	0.251	0.187	0.140	0.230	0.834	0.415

6.8 Table 8: identity and religiosity

Read:

- Panel (a) studies identity proxied by language outcomes.
- SD INPRES exposure does not increase national language use at home but increases Arabic literacy.
- Panel (b) studies Islamic piety and practice.
- Exposed cohorts are more likely to pray five times daily, read the Qur'an, engage in prayer practices, and score higher on the piety index.

Economic message: Rather than strengthening national identity, SD INPRES exposure strengthened religious identity and practice, especially among Muslims.

6.9 Table 9: intergenerational transmission

Read:

- Table 9 studies whether exposed parents transmit religious cultural investments to their children.
- Exposed fathers are more likely to use Arabic in the home and to have children with Arabic literacy.
- Exposed parents are more likely to sort into Islamic-educated partners or transmit Islamic education-related outcomes.

Economic message: The effects of mass public schooling did not end with the treated cohorts. Religious schooling and identity were transmitted across generations through household choices.

TABLE 9

SD INPRES exposure and religious cultural transmission

	Marriage matching		*Arabic literacy*			
	Islamic-educated partner		Arabic in the home Parents & children		Child's Arabic No Islamic schooling	
	(1)	(2)	(3)	(4)	(5)	(6)
INPRES × young (father)	0.0020**		0.0043*		0.0072**	
	(0.0009)		(0.0025)		(0.0036)	
INPRES × young (mother)		−0.0002		0.0048*		0.0054
		(0.0007)		(0.0026)		(0.0046)
Number of individuals	725,803	544,143	304,048	246,049	95,678	77,064
Number of districts	275	275	275	275	272	272
Dep. var. mean	0.039	0.024	0.213	0.268	0.877	0.887
R^2	0.038	0.026	0.112	0.138	0.048	0.043

7 Short summary

- SD INPRES was a massive public primary school construction programme in Indonesia in the 1970s.
- The paper studies how religious education providers responded to this state expansion.
- INPRES schools were built in areas with unmet schooling demand, including many areas with Islamic schools.
- Islamic schools entered more in high-INPRES areas after the programme.
- Entry was strongest where local funding capacity was greater, especially rice-rich and waqf-rich villages.
- New Islamic schools did not secularize; many became more religiously differentiated.
- SD INPRES reduced elementary madrasa attendance but increased secondary madrasa attendance.
- The programme did not increase electoral support for the secular regime.
- Exposed cohorts show stronger religious identity and Islamic piety.

- Religious identity effects transmitted intergenerationally.
- The main lesson is that state provision can strengthen rather than weaken competing religious public-goods systems when those systems can mobilize resources and differentiate.

8 Conclusion

8.1 Core conclusion

Bazzi, Hilmy, and Marx show that mass public schooling in Indonesia did not simply crowd out religious education. The state expansion changed the structure of schooling markets. Public primary schools absorbed primary enrollment, but Islamic schools expanded at the secondary level and became more religiously differentiated.

8.2 Economic intuition

The intuition is that public primary school construction created new cohorts of educated students who demanded further schooling. Islamic providers could enter downstream secondary markets, use local religious funding networks, and differentiate their curriculum from secular state schools. The state therefore expanded education while unintentionally strengthening a parallel religious education system.

8.3 Takeaway

The paper revises the standard view of nation building through schooling. Public education can promote development, but when it competes with organized religious providers, it can also trigger mobilization outside the state. The long-run result can be more schooling, but also stronger religious identity and a more durable dual system of public goods provision.